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Electronic device

Abstract

An electronic device includes a lower housing and an upper cover that are connected to each other. The upper cover has a bending structure that faces a direction of the lower housing, an end that is of the lower housing and that faces the upper cover has a flanging structure, the flanging structure overlaps a bottom wall of the bending structure, and the upper cover and the lower housing are connected at an overlapping part of the bending structure and the flanging structure.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to Chinese Patent Application No. 202222448927.4, filed on Sep. 14, 2022, which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

(2) The embodiments relate to the field of electronic technologies, and an electronic device.

BACKGROUND

(3) With continuous development and update of electronic technologies, electronic devices with energy storage functionality, for example, an inverter, a power distribution box, a switch box, and the like, are increasingly widely applied. An inverter is used as an example. The inverter is a converter that may convert direct current energy into a constant-frequency constant-voltage alternating current or a frequency-modulation voltage-regulation alternating current. The inverter includes a photovoltaic inverter. The photovoltaic inverter is an inverter that may convert a variable direct current generated by a photovoltaic solar panel into a mains frequency alternating current. The photovoltaic inverter may feed back the alternating current obtained through conversion to commercial transmission systems or to off-grid power grids for use.

(4) The foregoing electronic device may include a lower housing and an upper cover that covers the lower housing. The upper cover and the lower housing may be connected and fastened by using a fastener (for example, a bolt, a screw, a machine screw, or the like), to improve firmness and reliability of connection between the upper cover and the lower housing. A screw fastening manner is used as an example. A mounting hole may be provided on a front surface or a side surface of the

upper cover, and a threaded hole opposite the mounting hole is provided on the lower housing. In a connection process, a screw fastener may pass through the mounting hole of the upper cover, and then be screwed into the threaded hole of the lower housing, so that the upper cover and the lower housing are connected through the screw fastener.

(5) However, in the foregoing connection manner, the screw fastener is located on the front surface or the side surface of the upper cover; consequently, an appearance effect of the electronic device is relatively poor, and overall aesthetics of the electronic device is reduced.

SUMMARY

(6) The embodiments may provide an electronic device to resolve a problem that an appearance of the electronic device is relatively poor and overall aesthetics of the electronic device is reduced because a connection manner between an upper cover and a lower housing causes a screw fastener to be located on a front surface or a side surface of the upper cover in an existing electronic device.

(7) An electronic device includes a lower housing and an upper cover that are connected to each other.

(8) The upper cover has a bending structure that faces a direction of the lower housing, an end that is of the lower housing and that faces the upper cover has a flanging structure, the flanging structure overlaps a bottom wall of the bending structure, and the upper cover and the lower housing are connected at an overlapping part of the bending structure and the flanging structure.

(9) The upper cover and the lower housing are connected at an overlapping part of the flanging structure and the bottom wall of the bending structure, so that a connection structure between the upper cover and the lower housing may be hidden at the bottom of the upper cover. This effectively avoids occurrence of the connection structure on a side surface or a front surface of the upper cover, and avoids impact on an appearance of the electronic device caused by the occurrence of the connection structure on the front surface or the side surface of the upper cover, thereby effectively improving external integrity of the electronic device, and improving overall aesthetics of the electronic device.

(10) In addition, the upper cover has the bending structure, and the bending structure may cause the upper cover to deform when an inner part of the electronic device is affected by an energy shock, to increase a volume of a cavity inside the electronic device, thereby providing buffer space for the energy shock, effectively avoiding a crack between the upper cover and the lower housing, avoiding occurrence of a security accident, and effectively improving security of the electronic device.

(11) In a possible implementation, a fastening assembly is further included. The bottom wall of the bending structure and the flanging structure are connected at the overlapping part by using the fastening assembly. A connection manner structure of the fastening assembly is simple. This may simplify a connection on the upper cover and the lower housing. In addition, the fastening assembly provides reliable connection and is easy to disassemble. This may effectively improve reliability and firmness of connection between the upper cover and the lower housing, and may further facilitate disassembly between the upper cover and the lower housing, thereby facilitating repair and maintenance of the electronic device.

(12) In a possible implementation, the fastening assembly includes a bolt component and a nut component that fit with each other.

(13) A first assembly hole is provided on the bottom wall of the bending structure, a second assembly hole opposite to the first assembly hole is provided on the flanging structure, and a stud of the bolt component passes through the first assembly hole and the second assembly hole and fits with the nut component.

(14) One of the bolt component and the nut component is fastened to an inner side of the bottom wall of the bending structure. In this way, when the nut component and the bolt component are fastened and fitted, only one of the fastener needs to be rotated to implement fastening and fitting of the nut component and the bolt component, and no auxiliary fastening is required for a fastening

assembly fastened to the inner side of the bottom wall. This effectively reduces operation difficulty of fitting the fastening assemblies, improves tightness of fitting the fastener, and improves firmness and reliability of connection between the upper cover and the lower housing.

(15) In a possible implementation, the nut component is a self-clinching nut or a rivet nut, and the nut component is fastened to the inner side of the bottom wall and is coaxially disposed with the first assembly hole.

(16) The stud of the bolt component sequentially passes through the second assembly hole and the first assembly hole and fits with the nut component. In this way, in a connection process, the stud of the bolt component is extended into the nut component and rotated, so that the bolt component and the nut component may be fastened and fitted, and the nut component does not need to be fastened. This effectively avoids rotation of the nut component with the bolt component in a process of fitting with the bolt component, and avoids affecting normal fitting between the nut component and the bolt component caused by the rotation of the nut component with the bolt component, thereby effectively improving tightness of fitting the nut component and the bolt component, and improving firmness and reliability of connection between the upper cover and the lower housing.

(17) In addition, fastened connection operations between the bending structure and each of the self-clinching nut and the rivet nut are simple, and the connections are reliable and cause no damage or harm to an overall structure of the upper cover. This can effectively improve firmness and reliability of a connection between the nut component and the bottom wall, thereby improving reliability and stability of connection between the upper cover and the lower housing.

(18) In a possible implementation, the bolt component is a self-clinching bolt or a rivet bolt, a head of the bolt component is fastened to the inner side of the bottom wall, and the stud of the bolt component sequentially passes through the first assembly hole and the second assembly hole and extends out of an outer side of the flanging structure, to fit with the nut component. In this way, in a connection process, the nut component may be directly sleeved on the stud of the bolt component and the nut component is rotated, so that the nut component and the bolt component may be fastened and fitted, and the bolt component does not need to be fastened. This effectively avoids rotation of the bolt component in a process of fitting with the nut component, and avoids affecting normal fitting between the bolt component and the nut component caused by the rotation of the bolt component with the nut component, thereby effectively improving tightness of fitting the bolt component and the nut component, and improving firmness and reliability of connection between the upper cover and the lower housing.

(19) In a possible implementation, a sealing element is further included. The sealing element is disposed between the upper cover and the lower housing. The sealing element may play a sealing role for connection between the upper cover and the lower housing, avoid entering of external stains such as water stains and oil stains into the cavity of the electronic device, and prevent the foregoing stains from entering the cavity to affect normal operation of the electronic device, thereby helping improve stability and security of operation of the electronic device.

(20) In a possible implementation, the flanging structure has a first flip surface parallel to the bottom wall of the bending structure, and the bottom wall of the bending structure is bonded to the first flip surface.

(21) The bending structure and the flanging structure are connected at an overlapping part of the first flip surface and the bottom wall. In this way, the first flip surface may be more closely attached to the bottom wall, a fitting and attachment degree between the bending structure and the flanging structure can be effectively improved, and a coaxiality between the first assembly hole and the second assembly hole can be improved, thereby facilitating fastening and fitting between the bolt component and the nut component, helping improve firmness and reliability of fitting between the nut component and the bolt component, and improving reliability and stability of connection between the upper cover and the lower housing.

(22) In a possible implementation, a side wall of the lower housing is flipped outwards to form the flanging structure.

(23) The lower housing and the upper cover are enclosed to form a cavity, the flanging structure is concave towards a direction of the cavity to form a recess part, and the recess part is configured to protrude outwards when an impact force is exerted inside the lower housing, to increase a volume of the cavity. In this way, buffer space can be provided for an energy shock in the cavity, thereby effectively reducing a pulling force at a connection part between the upper cover and the lower housing, effectively reducing or avoiding a crack between the upper cover and the lower housing, preventing a security accident caused by the crack between the upper cover and the lower housing, and effectively improving security of the electronic device.

(24) In a possible implementation, the flanging structure further includes a second flip surface connected to the first flip surface.

(25) The second flip surface is located on a side that is of the first flip surface and that is close to the side wall of the lower housing, and the second flip surface is located on a side that is of the first flip surface and that faces the cavity, to enable the second flip surface to be concave towards the cavity to form the recess part.

(26) In a possible implementation, the sealing element is located between the second flip surface and the upper cover. The second flip surface is located on a side that is of the first flip surface and that is close to the upper cover, so that a distance between the second flip surface and the upper cover is relatively small. In this way, a compression amount of the sealing element may be effectively increased, and a sealing effect of connections between the sealing element and each of the upper cover and the lower housing is improved, thereby effectively improving sealing performance between the upper cover and the lower housing. In addition, this further helps improve firmness and reliability of setting the sealing element, effectively reduce or avoid deviation of the sealing element, and improve reliability and stability of sealing between the upper cover and the lower housing.

(27) In a possible implementation, the flanging structure is enclosed around a periphery of a side wall of the lower housing. The sealing element is clamped at an end that is of the side wall of the lower housing and that faces the upper cover.

(28) In a possible implementation, the electronic device is a photovoltaic inverter, which includes module terminals, a circuit board, power components, a fan, and a lower shell and an upper cover connected with each other. The power components and the fan are installed on the circuit board. The circuit board is mounted on a side wall or bottom wall of the lower housing. The module terminals are disposed on a side wall or a bottom wall of the lower housing, a part of the module terminals located outside the lower housing is configured to connect to a photovoltaic module, and a part of the module terminals located inside the lower housing is configured to connect to the circuit board and transmitting power of the photovoltaic module to the power component through the circuit board. The upper cover has a bending structure facing the lower housing, and an end of the lower housing facing the upper cover has a flanging structure, and the flanging structure overlaps a bottom wall of the bending structure. The upper cover is connected to the lower housing at a position at which the bending structure overlaps the flanging structure, so that a sealed accommodating cavity is formed between the upper cover and the lower housing, and the heat dissipation fan forms internal circulating heat dissipation air in the sealed accommodating cavity so as to evenly distribute heat generated by the power components in the sealed accommodation cavity, and prevent dust outside the cavity from entering the sealed accommodation cavity and affecting the power components.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic diagram of a structure of an electronic device according to an embodiment;
- (2) FIG. 1a is a schematic diagram of a structure of connection between an electronic device and a connector according to an embodiment;
- (3) FIG. 1b is a schematic diagram of an application scenario of an electronic device according to an embodiment;
- (4) FIG. 2 is a schematic exploded view of an electronic device according to an embodiment;
- (5) FIG. 3 is a sectional view of an electronic device according to an embodiment;
- (6) FIG. 4 is an enlarged view of an area A in FIG. 2;
- (7) FIG. 5 is an enlarged view of an area B in FIG. 3;
- (8) FIG. 5a is a schematic diagram of a partial structure of a sealing element disposed on an upper cover according to an embodiment;
- (9) FIG. 6 is a sectional view of another electronic device according to an embodiment; and
- (10) FIG. 7 is an enlarged view of an area C in FIG. 6.

DETAILED DESCRIPTION OF THE EMBODIMENTS

- (11) Terms used in the embodiments are only used to explain but are not intended as limiting.
- (12) For ease of understanding, related terms in the embodiments are first explained and described.
- (13) FIG. 1 is a schematic diagram of a structure of an electronic device according to an embodiment. FIG. 1a is a schematic diagram of a structure of a connection between the electronic device and a connector according to this embodiment.
- (14) This embodiment provides the electronic device. The electronic device may be an inverter, for example, a photovoltaic inverter, or the electronic device may be an electronic device configured to store energy, for example, a power distribution box, a switch box, or the like. For example, the electronic device may be configured to store electric energy, thermal energy, light energy, and the like. The following uses an example in which the electronic device is a photovoltaic inverter for description.
- (15) As shown in FIG. 1, the electronic device **100** may include a lower housing **110** and an upper cover **120** that are connected to each other. The lower housing **110** and the upper cover **120** may be enclosed to form a sealed cavity **150** (as shown in FIG. 3). A mechanical part configured to implement a complete function of the electronic device **100** may be disposed in the cavity **150**. For example, a mechanical part such as a circuit board, an inductor, a relay, a fuse, a power switch, a heat sink, or the like may be disposed in the cavity **150**, so that the electronic device **100** may implement the function of the electronic device **100** by using the foregoing mechanical part. For example, when the electronic device **100** is a photovoltaic inverter, further referring to FIG. 1 and FIG. 1a and FIG. 1b, the photovoltaic inverter **100** includes a lower housing **110** and an upper cover **120** that are connected to each other that are shown in FIG. 1. Further, as shown in FIG. 1a and FIG. 1b, the photovoltaic inverter **100** further includes a module terminals **180**, power components **191**, a circuit board **192**, and a fan **193**. The power components **191** and the fan **193** are mounted on the circuit board **192**, and the circuit board **192** is accommodated and mounted and fixed in the lower housing **110**. The module terminals **180** are disposed on a side wall or a bottom wall of the lower housing **110**, a part of the module terminals **180** located outside the lower housing **110** is configured to connect the photovoltaic module **200**, and a part of the module terminals **180** located inside the lower housing **110** is configured to connect the circuit board **192**. The power of the photovoltaic module **200** is transmitted to the power components **191** through the circuit board **192**. The upper cover **120** has a bending structure facing the lower housing **110**, an end of the lower housing **110** facing the upper cover **120** has a flanging structure **111**, and the flanging structure **111** overlaps a bottom wall of the bending structure **121**. The upper cover **120** is connected to the lower housing **110** at a position at which the bending structure **121** overlaps the flanging structure **111**, so

that a sealed accommodating cavity **150** is formed between the upper cover **120** and the lower housing **110**, so that the fan **193** forms internal circulation heat dissipation air in the sealed accommodating cavity **150** so that heat generated by the power components **191** is evenly distributed in the sealed accommodation cavity **150** and dust outside the cavity is prevented from entering the sealed accommodation cavity **150** to affect the power components **191**.

(16) The lower housing **110** and the upper cover **120** are configured to carry the foregoing mechanical part, to centralize the foregoing mechanical part in the cavity **150**, thereby improving structural stability and reliability of the electronic device **100**. The upper cover **120** and the lower housing **110** may be sheet metal parts. For example, the upper cover **120** and the lower housing **110** may be sheet metal parts formed through bending, welding, or another process. The sheet metal part is relatively light and thin and has a relatively low cost, which helps reduce a weight of the electronic device **100** and reduce costs of the electronic device **100**. In addition, the sheet metal part further has relatively good bending performance, which may facilitate bending of the lower housing **110** and the upper cover **120**, so that the lower housing **110** and the upper cover **120** can meet different shape requirements.

(17) As shown in FIG. **1a**, a connector (for example, a first connector **160**, a second connector **170**, or the like in FIG. **1a**) may be further disposed on the electronic device **100**. One end of the connector may be connected to the electronic device **100**, and the other end may be configured to be connected to an external device, so that the electronic device **100** may be connected to the external device by using the connector, to perform energy exchange, transmission, and the like.

(18) Currently, an upper cover and a lower housing of an electronic device may be connected by using a screw fastener. The upper cover and the lower housing with a sheet metal structure are relatively light and thin. In a process of connecting the upper cover and the lower housing by using the screw fastener, a mounting hole may be provided on a front surface or a side surface of the upper cover, a threaded hole opposite to the mounting hole may be provided on the lower housing, and the screw fastener may pass through the mounting hole and be screwed into the threaded hole, so that the upper cover and the lower housing may be connected by using the screw fastener. To improve firmness of connection between the upper cover and the lower housing, a plurality of screw fasteners may be disposed between the upper cover and the lower housing.

(19) However, in the foregoing connection manner, the screw fastener is located on the front surface or the side surface of the upper cover; consequently, an appearance effect of the electronic device is relatively poor, and aesthetics of the electronic device is reduced.

(20) In a related technology, to avoid occurrence of the screw fastener on the front surface or the side surface of the upper cover, the upper cover and the lower housing adopt die casting parts. The die casting part is thick. In this way, a threaded hole may be provided on a surface that is of the upper cover and that faces the lower housing (that is, the bottom of the upper cover), a through hole may be provided on the lower housing, and then the screw fastener extends into the threaded hole of the upper cover in a direction of the lower housing, to implement connection between the upper cover and the lower housing. Therefore, the screw fastener is hidden at the bottom of the upper cover, to avoid the occurrence of the screw fastener on the front surface of the upper cover.

(21) However, hardness and brittleness of the upper cover and the lower housing of a structure of the die casting part are relatively high. When an unexpected situation such as a short circuit occurs in a use process of the electronic device, and a cavity of the electronic device generates an energy shock, plastic deformation is difficult to occur because the hardness and brittleness of the upper cover and the lower housing are relatively high. In this way, the upper cover is prone to fall from the lower housing under an action of the energy shock in the cavity, and even cracks, consequently causing a security accident. Security of the electronic device is greatly reduced.

(22) Based on the foregoing problem, an embodiment provides an electronic device, in which a connection part between an upper cover and a lower housing may be located below the upper cover, thereby effectively avoiding impact on an appearance on the electronic device caused by

occurrence of a fastener on a front surface or a side surface of the upper cover, and effectively improving overall aesthetics of the electronic device.

(23) FIG. 2 is a schematic exploded view of an electronic device according to an embodiment. FIG. 3 is a sectional view of the electronic device according to this embodiment.

(24) In this embodiment, in the accompanying drawings, a direction x represents a length direction of the electronic device **100**, a direction y represents a width direction of the electronic device, and a direction z represents a thickness direction of the electronic device.

(25) The electronic device **100** provided in this embodiment is shown in FIG. 2. An upper cover **120** has a bending structure **121** that faces a direction of a lower housing **110**, and an end that is of the lower housing **110** and that faces the upper cover **120** has a flanging structure **111**. With reference to FIG. 3, the flanging structure **111** and a bottom wall of the bending structure **121** may overlap each other. For example, the flanging structure **111** and the bending structure **121** may overlap in the thickness direction (that is, the direction z in FIG. 3) of the electronic device **100**. The upper cover **120** and the lower housing **110** may be connected at an overlapping part of the bending structure **121** and the flanging structure **111**.

(26) For example, the bottom wall of the bending structure **121** may be bonded to the flanging structure **111**, so that the flanging structure **111** may provide support for the upper cover **120**. This helps provide auxiliary positioning and support for installation of the upper cover **120**, thereby improving accuracy and reliability of an installation operation between the upper cover **120** and the lower housing **110**.

(27) For example, the flanging structure **111** and the bottom wall of the bending structure **121** may be detachably connected in a manner of clamping, fastening, bolt connecting, or the like, so that the upper cover **120** is detachable from the lower housing **110**. This may facilitate maintenance of the electronic device **100**, and helps improve a service life of the electronic device **100**.

(28) As shown in FIG. 3, the upper cover **120** may include a top wall **122** and the bending structure **121** connected to the top wall **122**. The bending structure **121** may include a side wall **1211** and a bottom wall **1212** that are disposed at an angle. The side wall **1211** may be located between the top wall **122** and the bottom wall **1212**. An edge of the top wall **122** may be sequentially bent twice to form the side wall **1211** and the bottom wall **1212** of the bending structure **121**. For example, the edge of the top wall **122** may be bent for a first time along the thickness direction (that is, the direction z in FIG. 3) of the electronic device **100** and towards the lower housing **110** (that is, the direction -z in FIG. 3), to form the side wall **1211** of the bending structure **121**. Then, the side wall **1211** is bent for a second time along the length direction (the direction x shown in FIG. 3) of the electronic device **100** and towards a side wall of the lower housing **110**, to form the bottom wall **1212** of the bending structure **121**, so that the upper cover **120** forms the bending structure **121** after being bent twice.

(29) When an accident (for example, a short circuit) occurs on a mechanical part inside the electronic device **100**, and an energy shock is generated inside a cavity **150** of the electronic device **100**, after the upper cover **120** is affected by the energy shock, angles between the side wall **1211** and each of the top wall **122** and the bottom wall **1212** may change under an impact force, to change a structural shape of the upper cover **120**. For example, the angles between the side wall **1211** and each of the top wall **122** and the bottom wall **1212** may increase, so that the bending structure **121** is unfolded. In this way, a volume of the cavity **150** of the electronic device **100** may be increased, and buffer space is provided for an energy shock in the cavity **150**, thereby reducing a pulling force at a connection part between the upper cover **120** and the lower housing **110**, avoiding a security accident caused by a crack between the upper cover **120** and the lower housing **110**, and helping improve security of the electronic device **100**.

(30) The upper cover **120** and the lower housing **110** are connected at an overlapping part of the flanging structure **111** and the bottom wall **1212** of the bending structure **121**, so that a connection structure between the upper cover **120** and the lower housing **110** may be hidden at the bottom of

the upper cover **120**. This effectively avoids occurrence of the connection structure on a side surface or a front surface of the upper cover **120**, and avoids impact on an appearance of the electronic device **100** caused by the occurrence of the connection structure on the front surface or the side surface of the upper cover **120**, thereby effectively improving external integrity of the electronic device **100**, and improving overall aesthetics of the electronic device **100**.

(31) Compared with the electronic device in the related technology, in the electronic device **100** in this embodiment, the upper cover **120** has the bending structure **121**, and the bending structure **121** may cause the upper cover **120** to deform when an inner part of the electronic device **100** is affected by the energy shock, to increase the volume of the cavity **150** inside the electronic device **100**, thereby providing buffer space for the energy shock, effectively avoiding a crack between the upper cover **120** and the lower housing **110**, avoiding occurrence of a security accident, and effectively improving security of the electronic device **100**. In addition, a structure for enhancing structural strength of the upper cover **120** and the lower housing **110** that is disposed in the electronic device **100** may be omitted, so that the structure of the upper cover **120** and the lower housing **110** may be effectively simplified, and a cost of the electronic device **100** may be reduced.

(32) FIG. 4 is an enlarged view of an area A in FIG. 2.

(33) As shown in FIG. 4, the electronic device **100** may further include a fastening assembly **130**. The bottom wall **1212** of the bending structure **121** and the flanging structure **111** may be connected at the overlapping part by using the fastening assembly **130**. For example, the fastening assembly **130** may be a bolt fastening assembly, a screw fastening assembly, or the like. A connection manner of the fastening assembly **130** is simple. This may simplify a connection on the upper cover **120** and the lower housing **110**. In addition, the fastening assembly **130** provides reliable connection and is easy to disassemble. This may effectively improve reliability and firmness of connection between the upper cover **120** and the lower housing **110**, and may further facilitate disassembly between the upper cover **120** and the lower housing **110**, thereby facilitating repair and maintenance of the electronic device **100**.

(34) Still as shown in FIG. 4, for example, the fastening assembly **130** may include a bolt component **131** and a nut component **132** that fit with each other. A first assembly hole **1213** may be provided on the bottom wall **1212** of the bending structure **121**, a second assembly hole **1111** opposite to the first assembly hole **1213** may be provided on the flanging structure **111**, and a stud of the bolt component **131** may pass through the first assembly hole **1213** and the second assembly hole **1111** and fits with the nut component **132**, so that the upper cover **120** and the lower housing **110** may be connected through fitting between the bolt component **131** and the nut component **132**. One of the bolt component **131** and the nut component **132** may be fastened to an inner side of the bottom wall **1212** of the bending structure **121**.

(35) For example, the nut component **132** may be fastened to the inner side of the bottom wall **1212**, and then the stud of the bolt component **131** sequentially passes through the second assembly hole **1111** and the first assembly hole **1213**, to fit with the nut component **132**, so that the upper cover **120** and the lower housing **110** are connected through fitting between the bolt component **131** and the nut component **132**. Alternatively, the bolt component **131** may be fastened to the inner side of the bottom wall **1212**. For example, a head of the bolt component **131** may be fastened to the inner side of the bottom wall **1212**, and then the stud of the bolt component **131** sequentially passes through the first assembly hole **1213** and the second assembly hole **1111** and extends out of an outer side of the flanging structure **111**, so that the nut component **132** may fit with the protruding stud, and the upper cover **120** and the lower housing **110** may be connected through fitting between the bolt component **131** and the nut component **132**.

(36) One of the bolt component **131** and the nut component **132** is fastened to the inner side of the bottom wall **1212**. In this way, when the nut component **132** and the bolt component **131** are fastened and fitted, only one of the fasteners needs to be rotated to implement fastening and fitting of the nut component **132** and the bolt component **131**, and no auxiliary fastening is required for a

fastener fastened to the inner side of the bottom wall **1212**. This effectively reduces operation difficulty of fitting the fastening assembly **130**, helps improve tightness of fitting the fastening assembly **130**, and improves firmness and reliability of connection between the upper cover **120** and the lower housing **110**.

(37) The bolt component **131** and the nut component **132** may be fastened to the inner side of the bottom wall **1212** of the bending structure **121** in a manner of welding, bonding, clamping, or fastening, or the bolt component **131** and the nut component **132** may be fastened to the inner side of the bottom wall **1212** in another connection manner.

(38) For example, in a possible implementation, the nut component **132** may be a self-clinching nut or a rivet nut, and the nut component **132** may be fastened to the inner side of the bottom wall **1212** and disposed coaxially with the first assembly hole **1213**. The self-clinching nut is used as an example. One end of the self-clinching nut has an embossing tooth, and the end of the self-clinching nut having the embossing tooth is pressed into the bottom wall **1212** of the flanging structure **111**, so that plastic deformation may occur on the bottom wall **1212**, and the self-clinching nut may be connected to the bottom wall **1212** through fastening by using the embossing tooth. In this case, the stud of the bolt component **131** may sequentially pass through the second assembly hole **1111** and the first assembly hole **1213**, and extend into the self-clinching nut fastened to the inner side of the bottom wall **1212**. The bolt component **131** and the nut component **132** may be fastened and fitted by rotating the bolt component **131**, so that the upper cover **120** and the lower housing **110** may be connected through fitting between the bolt component **131** and the nut component **132**.

(39) Correspondingly, a use principle of the rivet nut is similar to a use principle of the self-clinching nut. For example, the rivet nut is pulled out to connect the rivet nut and the bottom wall **1212** through fastening, so that the stud of the bolt component **131** may sequentially pass through the second assembly hole **1111** and the first assembly hole **1213**, to fit with the self-clinching nut fastened to the inner side of the bottom wall **1212**, so as to implement connection between the upper cover **120** and the lower housing **110**.

(40) The nut component **132** and the bottom wall **1212** of the bending structure **121** may be connected by using the self-clinching nut or the rivet nut, to fasten the nut component **132**. In this way, in a connection process, the stud of the bolt component **131** is extended into the nut component **132** and rotated, so that the bolt component **131** and the nut component **132** may be fastened and fitted, and the nut component **132** does not need to be fastened. This effectively avoids rotation of the nut component **132** with the bolt component **131** in a process of fitting with the bolt component **131**, and avoids affecting normal fitting between the nut component **132** and the bolt component **131** caused by the rotation of the nut component **132** with the bolt component **131**, thereby effectively improving tightness of fitting the nut component **132** and the bolt component **131**, and improving firmness and reliability of connection between the upper cover **120** and the lower housing **110**.

(41) In addition, fastened connection operations between the self-clinching nut and the bending structure **121**, and between the rivet nut and the bending structure **121** are simple, and the connections are reliable and cause no damage or harm to an overall structure of the upper cover **120**. This can effectively improve firmness and reliability of connection between the nut component **132** and the bottom wall **1212**, thereby improving reliability and stability of connection between the upper cover **120** and the lower housing **110**.

(42) In another possible implementation, the bolt component **131** may be a self-clinching bolt or a rivet bolt, and a head of the bolt component **131** may be fastened to the inner side of the bottom wall **1212** and disposed coaxially with the first assembly hole **1213**. The stud of the bolt component **131** may sequentially pass through the first assembly hole **1213** and the second assembly hole **1111** and extend out of an outer side of the flanging structure **111**, so that the nut component **132** may fit with the stud. The self-clinching bolt is used as an example. One end that is of the head of the self-

clinch bolt and that faces the stud has an embossing tooth. In a use process, the stud of the self-clinching bolt may sequentially pass through the first assembly hole **1213** and the second assembly hole **1111** from an inner side of the bottom wall **1212** to extend out of the outer side of the flanging structure **111**, and the end that is of the head of the self-clinching bolt and that has the embossing tooth is pressed into the bottom wall **1212** of the flanging structure **111**, so that plastic deformation occurs on the bottom wall **1212**, and the head of the self-clinching bolt may be fastened to the bottom wall **1212** by using the embossing tooth. In this case, the nut component **132** is sleeved on a part of the stud that extends out of the outer side of the flanging structure **111** in the self-clinching bolt, and is screwed, so that the nut component **132** can be fastened and connected to the bolt component **131**, and the upper cover **120** and the lower housing **110** may be connected through fitting between the nut component **132** and the self-clinching bolt.

(43) Correspondingly, a use principle of the rivet bolt is similar to a use principle of the self-clinching bolt. For example, the rivet bolt is pulled out to connect the head of the rivet bolt to the bottom wall **1212** through fastening, so that a stud part of the rivet bolt may extend out of the outer side of the flanging structure **111**, and the nut component **132** can be fastened and fitted with the stud part.

(44) The bolt component **131** may be fastened to the bottom wall **1212** of the bending structure **121** by using the self-clinching bolt or the rivet bolt. In this way, in a connection process, the nut component **132** may be directly sleeved on the stud of the bolt component **131** and the nut component **132** is rotated, so that the nut component **132** and the bolt component **131** may be fastened and fitted, and the bolt component **131** does not need to be fastened. This effectively avoids rotation of the bolt component **131** in a process of fitting with the nut component **132**, and avoids affecting normal fitting between the bolt component **131** and the nut component **132** caused by the rotation of the bolt component **131** with the nut component **132**, thereby effectively improving tightness of fitting the bolt component **131** and the nut component **132**, and improving firmness and reliability of connection between the upper cover **120** and the lower housing **110**.

(45) Still as shown in FIG. 4, the electronic device **100** may further include a sealing element **140**. The sealing element **140** may be disposed between the upper cover **120** and the lower housing **110**. The sealing element **140** may play a sealing role for connection between the upper cover **120** and the lower housing **110**, avoid entering of external stains such as water stains and oil stains into the cavity **150** of the electronic device **100**, and prevent the stains from entering the cavity **150** to affect normal operation of the electronic device **100**, thereby helping improve stability and security of the electronic device **100**.

(46) FIG. 5 is an enlarged view of an area B in FIG. 3. FIG. 5a is a schematic diagram of a partial structure of the sealing element disposed on the upper cover according to this embodiment.

(47) As shown in FIG. 5, in a possible implementation, the flanging structure **111** may have a first flip surface **1112** parallel to the bottom wall **1212** of the bending structure **121**, and the bottom wall **1212** of the bending structure **121** may be bonded to the first flip surface **1112**. The bending structure **121** and the flanging structure **111** may be connected at an overlapping part of the first flip surface **1112** and the bottom wall **1212**. For example, the second assembly hole **1111** on the flanging structure **111** may be provided on the first flip surface **1112**. In a connection process of the upper cover **120** and the lower housing **110**, the bottom wall **1212** of the bending structure **121** may be bonded to the first flip surface **1112** of the flanging structure **111**, and the first assembly hole **1213** is opposite to the second assembly hole **1111**, so that the stud of the bolt component **131** may sequentially pass through the first assembly hole **1213** and the second assembly hole **1111**.

(48) The first flip surface **1112** is parallel to the bottom wall **1212**, so that the first flip surface **1112** may be more closely attached to the bottom wall **1212**, a fitting and attachment degree between the bending structure **121** and the flanging structure **111** can be effectively improved, and a coaxiality between the first assembly hole **1213** and the second assembly hole **1111** can be improved, thereby facilitating fastening and fitting between the bolt component **131** and the nut component **132**,

helping improve firmness and reliability of fitting between the nut component **132** and the bolt component **131**, and improving reliability and stability of connection between the upper cover **120** and the lower housing **110**.

(49) Still as shown in FIG. 5, a side wall of the lower housing **110** may be flipped outwards to form the flanging structure **111**. The flanging structure **111** is concave towards a direction of the cavity **150** of the electronic device **100** to form a recess part **1113**, and the recess part **1113** may be configured to protrude outwards when an impact force is exerted inside the lower housing **110**, to increase the volume of the cavity **150** of the electronic device **100**. For example, when an accident occurs on a mechanical part inside the electronic device **100**, and an energy shock is generated inside the cavity **150**, the recess part **1113** on the flanging structure **111** may deform to protrude outwards under the impact force when affected by the energy shock. In this way, the volume of the cavity **150** of the electronic device **100** may be increased, and buffer space can be provided for an energy shock in the cavity **150**, thereby effectively reducing a pulling force at a connection part between the upper cover **120** and the lower housing **110**, effectively reducing or avoiding a crack between the upper cover **120** and the lower housing **110**, preventing a security accident caused by the crack between the upper cover **120** and the lower housing **110**, and effectively improving security of the electronic device **100**.

(50) Still as shown in FIG. 5, the flanging structure **111** may further include a second flip surface **1114** connected to the first flip surface **1112**. The second flip surface **1114** may be located on a side that is of the first flip surface **1112** and that is close to the side wall of the lower housing **110**, and the second flip surface **1114** may be located on a side that is of the first flip surface **1112** and that faces the cavity **150**, so that the second flip surface **1114** is concave towards the cavity **150** to form the recess part **1113**.

(51) For example, as shown in FIG. 5, a connection surface **1115** may be provided between the first flip surface **1112** and the second flip surface **1114**, and the first flip surface **1112** and the second flip surface **1114** may be connected through the connection surface **1115**. In a flipping process, the side wall of the lower housing **110** may be first flipped outwards (that is, a direction backward to the cavity **150** of the electronic device **100**), for example, may be flipped along the length direction of the electronic device **100** (that is, the direction x in FIG. 5) and backward to the cavity **150** of the electronic device **100** (that is, the direction $-x$ in FIG. 5), to form the second flip surface **1114**. Then, the second flip surface **1114** is flipped in a direction backward to the upper cover **120**, for example, may be flipped in the thickness direction of the electronic device **100** (that is, the direction z in FIG. 5) and backward to the upper cover **120** (that is, the direction $-z$ in FIG. 5), to form the connection surface **1115**, so that the connection surface **1115** is located on a side that is of the second flip surface **1114** and that is backward to the upper cover **120**. Finally, the connection surface **1115** is continuously flipped outwards to form the first flip surface **1112**. For example, a flipping direction of the connection surface **1115** may be parallel to a flipping direction of the side wall of the lower housing **110**, to form the first flip surface **1112**, so that the first flip surface **1112** is located on a side that is of the second flip surface **1114** and that is away from the cavity **150**, and the second flip surface **1114** may be concave to form a recess part **1113**.

(52) When an accident occurs in the electronic device **100**, and an energy shock is generated inside the cavity **150**, a position relationship between the first flip surface **1112**, the connection surface **1115**, and the second flip surface **1114** of the flanging structure **111** may change under an effect of the energy shock. The second flip surface **1114** may protrude outwards under an impact force, so that the second flip surface **1114** is located on a side that is of the first flip surface **1112** and that is away from the cavity **150**, to increase the volume of the cavity **150**, thereby achieving an energy buffering effect, and avoiding a crack between the upper cover **120** and the lower housing **110** under the effect of the energy shock.

(53) The sealing element **140** may be located between the second flip surface **1114** and the upper cover **120**. For example, the sealing element **140** may be disposed on the second flip surface **1114**

in a manner shown in FIG. 5. For example, the sealing element **140** may be disposed on the second flip surface **1114** in a manner of bonding, clamping, or fastening. The manner of bonding is used as an example. A bonding layer may be disposed on a surface that is of the sealing element **140** and that faces the second flip surface **1114**, and the sealing element **140** may be bonded to the second flip surface **1114** by using the bonding layer.

(54) Alternatively, as shown in FIG. 5a, the sealing element **140** may be further disposed on the upper cover **120**. For example, the sealing element **140** may be disposed on the upper cover **120** in a manner of bonding, clamping, or fastening. For example, a bonding layer may be disposed on a side that is of the sealing element **140** and that faces the upper cover **120**, and the sealing element **140** may be bonded to the upper cover **120** by using the bonding layer. In a process of disposing the sealing element **140** on the upper cover **120**, the sealing element **140** may be located at a part that is on the top wall **122** of the upper cover **120** and that is opposite to the second flip surface **1114**, so that the sealing element **140** may be clamped between the top wall **122** of the upper cover **120** and the second flip surface **1114** when the upper cover **120** is disposed to cover the lower housing **110**. Therefore, the sealing element **140** may be in a compressed state between the second flip surface **1114** and the top wall **122**. This improves a sealing effect between the upper cover **120** and the lower housing **110**, and helps improve waterproof performance of the electronic device **100**.

(55) The second flip surface **1114** is located on a side that is of the first flip surface **1112** and that is close to the upper cover **120**, so that a distance between the second flip surface **1114** and the upper cover **120** is relatively small. The sealing element **140** is disposed on the second flip surface **1114**, so that a compression amount of the sealing element **140** may be effectively increased, and a sealing effect of connection between the sealing element **140** and each of the upper cover **120** and the lower housing **110** may be effectively improved, thereby improving sealing performance between the upper cover **120** and the lower housing **110**. In addition, this further helps improve firmness and reliability of the sealing element **140**, effectively reduce or avoid deviation of the sealing element **140**, and improve reliability and stability of sealing between the upper cover **120** and the lower housing **110**.

(56) It should be noted that, the sealing element **140** in FIG. 5 is a structural state of the sealing element **140** when the sealing element **140** is not squeezed by the upper cover **120** and the lower housing **110**. The sealing element **140** is an elastic element. In an actual use process, elastic deformation occurs on the sealing element **140** under an extrusion effect of the upper cover **120** and the lower housing **110**, to provide a sealing effect for the electronic device **100**.

(57) FIG. 6 is a sectional view of another electronic device according to an embodiment. FIG. 7 is an enlarged view of an area C in FIG. 6.

(58) In another possible implementation, as shown in FIG. 6 and FIG. 7, a flanging structure **111** may be further enclosed around a periphery of a side wall of a lower housing **110**, and a sealing element **140** may be clamped at an end that is of the side wall of the lower housing **110** and that faces an upper cover **120**. For example, the flanging structure **111** may be disposed on the periphery of the side wall of the lower housing **110** in a welding manner, so that a bending structure **121** of the upper cover **120** may be connected to the lower housing **110** by using the flanging structure **111**.

(59) A clamping slot may be disposed on the sealing element **140**. The clamping slot may be clamped on the side wall of the lower housing **110**, so that the sealing element **140** may be fastened to the lower housing **110**, and the sealing element **140** may achieve a sealing effect between the upper cover **120** and the lower housing **110**. In this way, firmness and reliability of disposing the sealing element **140** on the lower housing **110** may be effectively improved, and fall-off of the sealing element **140** can be effectively avoided, thereby effectively improving reliability and stability of disposing the sealing element **140**, and improving sealing performance of the electronic device **100**.

(60) Still as shown in FIG. 7, the side wall of the lower housing **110** may protrude outwards, to

increase a volume of a cavity **150** of the electronic device **100**, thereby providing buffer space for an energy shock inside the electronic device **100**, reducing an impact force of the energy shock on the upper cover **120** and the lower housing **110**, and improving security of the electronic device **100**.

(61) It should be noted that, the sealing element **140** in FIG. **6** and FIG. **7** is also an initial structural state of the sealing element **140**. In an actual use process, the sealing element **140** undergoes elastic deformation between the upper cover **120** and the lower housing **110**, instead of being in the state in FIG. **6** and FIG. **7**.

(62) In descriptions of the embodiments, it should be noted that, unless otherwise clearly specified and limited, the terms “installation”, “connection to”, and “connection” should be understood in a broad sense. For example, the connection may be a fixed connection, may be an indirect connection by using an intermediate medium, or may be an internal connection between two elements or an interaction relationship between two elements. For persons of ordinary skill in the art, specific meanings of the foregoing terms in the embodiments may be understood based on a specific situation. The terms “first”, “second”, “third”, “fourth”, and the like (if any) are intended to distinguish between similar objects but do not necessarily indicate an order or sequence.

(63) Finally, it should be noted that the foregoing embodiments are merely intended for describing the embodiments rather than limiting the embodiments. Although described in detail with reference to the foregoing embodiments, persons of ordinary skill in the art should understand that they may still make modifications to without departing from the scope of the solutions of the embodiments.

Claims

1. A photovoltaic inverter, comprising: a lower housing; an upper cover, which has a bending structure that faces a direction of the lower housing, an end of which faces the upper cover has a flanging structure, the flanging structure overlaps a bottom wall of the bending structure, and the upper cover and the lower housing are connected at an overlapping part of the bending structure and the flanging structure, so that a sealed accommodating cavity is formed between the upper cover and the lower housing; a fan; power components; a circuit board, on which the fan and the power components are configured to be installed; module terminals, disposed on a side wall or a bottom wall of the lower housing, and a part of the module terminals located outside the lower housing is configured to connect to a photovoltaic module, and a part of the module terminals located inside the lower housing is configured to connect to the circuit board, and are configured to transmit power of the photovoltaic module to the power component through the circuit board; wherein the fan is configured to form internal circulating heat dissipation air in the sealed accommodating cavity so as to evenly distribute heat generated by the power components in the sealed accommodation cavity; and prevent dust outside from entering the sealed accommodation cavity and affecting the power components; and a bolt component and a nut component that fit with each other; a first assembly hole is provided on the bottom wall of the bending structure, a second assembly hole opposite to the first assembly hole is provided on the flanging structure, and a stud of the bolt component passes through the first assembly hole and the second assembly hole and fits with the nut component; and one of the bolt component and the nut component is fastened to an inner side of the bottom wall of the bending structure which is facing to the cover.

2. The photovoltaic inverter according to claim 1, wherein the nut component is a self-clinching nut or a rivet nut, and the nut component is fastened to the inner side of the bottom wall and is coaxially disposed with the first assembly hole; and the stud of the bolt component sequentially passes through the second assembly hole and the first assembly hole and fits with the nut component.

3. The photovoltaic inverter according to claim 1, wherein the bolt component is a self-clinching bolt or a rivet bolt, a head of the bolt component is fastened to the inner side of the bottom wall,

and the stud of the bolt component sequentially passes through the first assembly hole and the second assembly hole and extends out of an outer side of the flanging structure, to fit with the nut component.

4. The photovoltaic inverter according to claim 1, further comprising: a sealing element, wherein the sealing element is disposed between the upper cover and the lower housing.

5. The photovoltaic inverter according to claim 4, wherein the flanging structure has a first flip surface parallel to the bottom wall of the bending structure, and the bottom wall of the bending structure is bonded to the first flip surface; and the bending structure and the flanging structure are connected at an overlapping part of the first flip surface and the bottom wall.

6. The photovoltaic inverter according to claim 5, wherein a side wall of the lower housing is flipped outwards to form the flanging structure; and the lower housing and the upper cover are enclosed to form a cavity, the flanging structure is concave towards a direction of the cavity to form a recess part, and the recess part is configured to protrude outwards when an impact force is exerted inside the lower housing, to increase a volume of the cavity.

7. The photovoltaic inverter according to claim 6, wherein the flanging structure further comprises: a second flip surface connected to the first flip surface; and the second flip surface is located on a side that is of the first flip surface and that is close to the side wall of the lower housing, and the second flip surface is located on a side that is of the first flip surface and that faces the cavity, to enable the second flip surface to be concave towards the cavity to form the recess part.

8. The photovoltaic inverter according to claim 7, wherein the sealing element is located between the second flip surface and the upper cover.

9. The photovoltaic inverter according to claim 5, wherein the flanging structure is enclosed around a periphery of a side wall of the lower housing; and the sealing element is clamped at an end that is of the side wall of the lower housing and that faces the upper cover.
